Inequalities and Tail Bounds

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Using the structure of the provided example scribe, generate a lecture scribe for the attached lecture slides. The scribe must serve as exam reference material for CSE 400: Fundamentals of Probability in Computing, Lecture 21 and 22: Inequalities and Tail Bounds. Include all definitions, notation, assumptions, theorem statements, complete proofs or derivations, and worked examples exactly as they appear in the provided lecture slides. Do not introduce any content, examples, explanations, or intuitions not present in the provided materials. Organize the scribe with clear numbered sections following the lecture structure: Tail and Motivation of Tail Bound, Markov's Inequality (definition, proof, example), Chebyshev's Inequality (definition, proof, example), Chernoff Bound and Poisson Tail (definition, proof, example), Jensen's Inequality (convex functions, proof, example), and Case Study and System Level Understanding. Prioritize correctness and completeness so that a student studying only this document can reliably revise everything taught in the lecture.

Below is a **formal lecture scribe** for *CSE 400: Fundamentals of Probability in Computing — Lecture 21 & 22 (Inequalities and Tail Bounds)*, strictly constructed from the provided slides and following the structure of the given template.

CSE 400 Lecture Scribe

Lecture 21 & 22: Inequalities and Tail Bounds

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1. Tail and Motivation of Tail Bounds

1.1 Definition of Tail

The **tail** of a random variable X is:

$$P\{X > x\}$$

1.2 Why Do We Care About Tails?

- Jobs delayed more than 24 hours
- Packets dropped due to full buffer

Key Idea:

- Mean → average behavior (easy)
- Tail → rare extreme events (hard)

1.3 Why Are Tails Hard to Compute?

- Requires summing many small probabilities

1.4 Tail Examples

Example 1: Load Balancing

- Distribute n jobs among n servers randomly
- Let $X \sim \text{Binomial}(n, p)$
- Tail probability:

$$P\{X > k\} = \sum \binom{n}{i} p^i (1-p)^{n-i}$$

This is a **tail event (rare overload)**.

Example 2: Poisson Arrivals

- Jobs arrive with rate λ
- $X \sim \text{Poisson}(\lambda)$

Tail probability:

$$P\{X > k\} = \sum e^{-\lambda} \frac{\lambda^x}{x!}$$

This represents a **tail event (sudden spike in load)**.

1.5 Why Tail Bounds?

- Exact computation is complicated:

$$P\{X > k\} = \sum (\text{long sums})$$

Idea:

- Instead of exact value → find an **upper bound**
- Easier to compute
- Still gives a guarantee

1.6 Running Example

- $X \sim \text{Binomial}(n, 1/2)$
- Mean:

$$E[X] = \frac{n}{2}$$

Question:

$$P(X > 3n/4)$$

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2. Markov's Inequality

2.1 Definition

For non-negative random variable X:

$$P(X > a) \leq \frac{E[X]}{a}$$

2.2 Key Idea

- Uses only the mean
- Large values are rare

2.3 Proof

$$E[X] = \sum_x x p_X(x)$$

Consider only $x > a$:

$$E[X] \geq \sum_{x>a} x p_X(x)$$

Replace x by a:

$$E[X] \geq \sum_{x>a} a p_X(x) = aP(X > a)$$

Thus:

$$P(X > a) \leq \frac{E[X]}{a}$$

2.4 Example (Income Distribution)

- Average income = 10,000
- Threshold = 50,000

$$P(X > 50,000) \leq \frac{10,000}{50,000} = 0.2$$

At most 20% people can earn 50,000

2.5 Solved Example 1 (Exam Grades)

- $E[X] = 70$, threshold $a = 90$

$$P(X > 90) \leq \frac{70}{90}$$

Result:

$$P(X > 90) < 0.778$$

2.6 Solved Example 2 (Server Load)

- $E[X] = 5000$, $a = 20000$

$$P(X > 20000) \leq \frac{5000}{20000} = 0.25$$

At most 25% chance

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3. Chebyshev's Inequality

3.1 Definition

$$P(|X - \mu| > a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

3.2 Key Idea

- Uses mean + variance
- Controls deviation from mean

3.3 Proof

Apply Markov to $(X - \mu)^2$:

$$P(|X - \mu| > a) = P((X - \mu)^2 > a^2)$$

$$\leq \frac{E[(X - \mu)^2]}{a^2} = \frac{\mathrm{Var}(X)}{a^2}$$

3.4 Running Example

- $X \sim \text{Binomial}(n, 1/2)$

$$P(X > 3n/4) \leq P(|X - n/2| > n/4)$$

Bound decreases as n increases

3.5 Real-Life Example

- Mean = 70, std = 10
- Deviation = 30

$$P(|X - 70| > 30) \leq \frac{10^2}{30^2} = \frac{100}{900} = \frac{1}{9}$$

At most $\frac{1}{9}$ students are far from mean

3.6 Solved Example

- $\mu = 70, \sigma = 5 \Rightarrow \sigma^2 = 25$
- $X > 90 \Rightarrow |X - 70| > 20$

$$P(X > 90) \leq \frac{25}{20^2}$$

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4. Chernoff Bound and Poisson Tail

4.1 Key Idea

- Uses exponential transformation
- Stronger than Markov and Chebyshev
- Exponential decay of tail

4.2 Core Trick

$$P(X > a) = P(e^{tX} > e^{ta})$$

Apply Markov:

$$P(X > a) \leq \frac{E[e^{tX}]}{e^{ta}}$$

4.3 Final Form

$$P(X > a) \leq \min_{t>0} \frac{E[e^{tX}]}{e^{ta}}$$

4.4 Lower Tail

$$P(X < a) \leq \min_{t<0} \frac{E[e^{tX}]}{e^{ta}}$$

4.5 Poisson Tail Bound

For $X \sim \text{Poisson}(\lambda)$:

$$E[e^{tX}] = e^{\lambda(e^t - 1)}$$

$$P(X > a) < e^{\lambda(e^t - 1) - at}$$

4.6 Solved Example 1 (Binomial Lower Tail)

- $X \sim \text{Binomial}(20, 0.5)$, $\mu = 10$
- Want $P(X < 0)$

Steps:

1. $0 = (1 - \delta)\mu \Rightarrow \delta = 1$
2. Use bound:

$$P(X < (1-\delta)\mu) < e^{-\mu\delta^2/2}$$

3. Substitute:

$$P(X < 0) < e^{-10/2} = e^{-5}$$

Result:

$$P(X < 0) < 0.0067$$

4.7 Solved Example 2 (Derivation)

Steps:

1. Choose $t < 0$:

$$P(X < a) = P(tX > ta)$$

2. Exponentiate:

$$P(e^{tX} > e^{ta})$$

3. Apply Markov:

$$\leq \frac{E[e^{tX}]}{e^{ta}}$$

4. Minimize over $t < 0$

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5. Jensen's Inequality

5.1 Convex Functions

A function g is convex if:

$$g''(x) > 0$$

5.2 Statement

For convex g:

$$g(E[X]) \leq E[g(X)]$$

5.3 Example (Verification)

Given:

- $X = 2$ w.p. 0.5, $X = 4$ w.p. 0.5
- $g(x) = x^2$

Steps:

1. $E[X] = 3$
2. $g(E[X]) = 9$
3. $E[g(X)] = 0.5(4) + 0.5(16) = 10$

Result:

$$g(E[X]) = 9 < 10 = E[g(X)]$$

5.4 Application

$$E[X^a] \geq (E[X])^a$$

for $a > 1$

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6. Case Study and System-Level Understanding

6.1 Problem Setting

- Demand X: requests/sec
- Capacity C
- Failure event:

$$X > C$$

Requirement:

$$P(X > C) < 0.01$$

6.2 Applying Inequalities

(a) Markov

- Only mean known ($\mu = 10,000$)
- Result:

$$C = 1,000,000$$

(b) Chebyshev

- Mean = 10,000, std = 2,000

$$\frac{2000^2}{(C - 10000)^2} < 0.01$$

Result:

$$C = 30,000$$

(c) Chernoff

- Full distribution known (Poisson)

$$P(X > C) < e^{\cdots}$$

Solve:

$$C = 14,500$$

6.3 Final Insight

Inequality	Information Used	Capacity
Markov	Mean only	1,000,000
Chebyshev	Mean + Variance	30,000
Chernoff	Full distribution	14,500

6.4 Golden Rule

> More statistical information = Tighter bounds = Lower hardware costs

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End of Scribe